

HCAI5TML01 – Mathematics of Learning.
Week – 3: Lecture – 01 – “Dice, Data & Density”
Probability and Generalization in Learning.
A **Refresher** on the **Probability, Random Variable** and it’s
Distribution.

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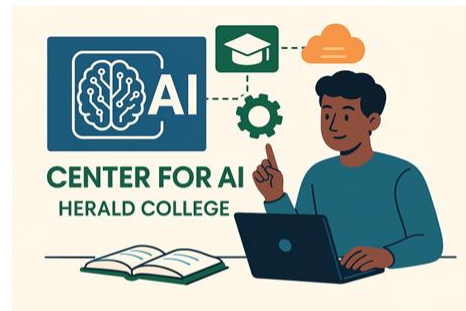


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1. Towards Understanding : Commonsense that is Probability.

{Understanding Probability Theory: With in Axiomatic Framework of Kolmogorov.}



1.1 What is Probability Theory?

- “*Probability theory is nothing but common sense reduced to calculation.*”
— *Pierre Laplace, 1812*
- Towards Probability Theory ...
 - In general, **Probability** is an **estimate or quantification of uncertainty** attached to an **event** related to some **process**.
 - Where do the **Uncertainty** may arise from:
 - **Noisy Measurements** in data collection,
 - **Natural Variability** between samples or individuals,
 - **Limited data** or finite sample sizes,
 - Probability provides a **consistent framework** for the quantification and manipulation of **uncertainty**.
 - In order to **model the behavior** of a **process** based on **observed** or **empirical outcomes** and make **inferences** about **future events**,
 - we adopt a formal, mathematical interpretation of probability – grounded in the axioms introduced by Andrey Kolmogorov (1933).

1.2 Foundation of Probability Theory

- **Axiomatic or Mathematical Interpretation:**

- We have **multiple interpretations of probability** – Classical, Empirical (frequentist), and Subjective (Bayesian) each **offering a different perspective** on **how to assign and reason about uncertainty**.
 - While these interpretations differ in philosophy and application, they all rely on a common **mathematical foundation of Probability Theory**.
- The axiomatic interpretation, developed by **Andrey Kolmogorov**, treats probability as a **mathematical function** defined on a **set of outcomes** (a **sample space**).
- His **axiomatic system** defines **probability as a function** satisfying specific logical rules (non – negativity, normalization, and additivity).
- This rigorous structure allows us to study **all interpretations** within a **unified theory**,
 - enabling clear reasoning, consistent modeling, and broad applicability
 - from games of chance to weather forecasting and machine learning.



A. Kolmogorov

1.3 A Probabilistic Space.

- A probability space is a formal **Mathematical Construct** that models a random experiment and consists of:
 - a random phenomenon {experiment},
 - defined by its **sample space**,
 - **outcomes within the sample space**,
 - and **probability measure**
 - {defines how to assign probability to each event governed by Axioms of Probability}.

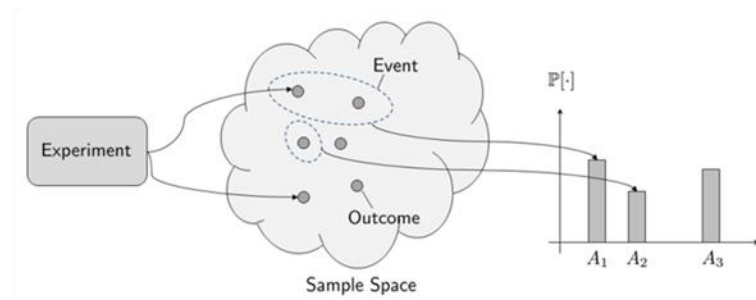


image by: S.Chan from Introduction to Probability for Datascience.

Fig: Elements of Probabilistic Space

-  **Probabilistic Space – A Formal Triplet Notation:**
 - A probability space is denoted as $(\Omega, \mathcal{F}, \mathbf{P})$:
 - Ω : sample space
 - $\mathcal{F}$ (or $\mathcal{A}$): σ -algebra i.e. collection of events; a set system closed under complements and countable unions).
 - $\mathbf{P}$: Probability Measure.

1.3.1 A {Chance} Experiment.



- {Chance} Experiment aka Random Process:
 - An experiment or process that leads to **uncertain outcomes (observation)**, even though it may be repeatable under the same conditions.
 - A chance experiment or random process by which an **observation** is made is called **Trial**.
 - Key **Features** of **Chance Experiment**:
 - **Reproducibility**: The experiment can be **conducted multiple times** under identical or controlled conditions.
 - **Uncertainty**: The **exact outcome** of any **single trial** cannot be **determined in advance**.
 - **Sample Space**: The set of all **possible outcomes**, known as the **sample space**, is **well-defined**.
- The **chance experiment** provides the context or process that gives rise **to the sample space** and, by extension, **the probability space**.

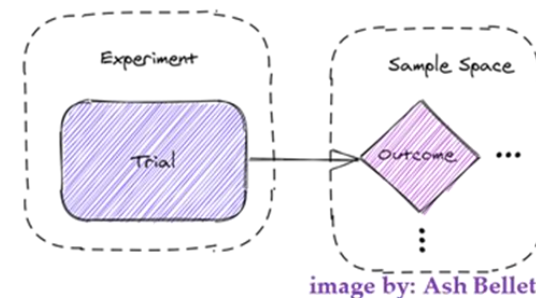


image by: Ash Bellet

1.3.2 Elements of Probabilistic Space.

- A **probability space** is a mathematical triplet $(\Omega, \mathcal{F}, \mathbf{P})$ that provides the formal framework for probability theory.

1. A Sample Space (Ω or $\mathbf{S}$):

- A **sample space** is the **set** of all **possible outcomes** of a **chance experiment**.
 - It is usually denoted by Ω (read as omega) or a capital letter $\mathbf{S}$. { Symbols may be used interchangeably. }
- **Types of Sample Space:**
 - **Discrete Sample Space:**
 - Contains a **finite or countably infinite** set of outcomes.
 - For example:
 - **Finite Example: Tossing a Coin $\rightarrow \Omega = \{\text{Heads, Tails}\}$**
 - **Countably Infinite Example: Rolling a die until a 6 appears $\rightarrow \Omega = \{1, 2, 3, \dots\}$**
 - **Continuous Sample Space:**
 - Contains an **uncountably infinite set of outcomes**, often represented as **intervals** on $\mathbb{R}$ or $\mathbb{R}^n$.
 - Common in measurements like time, position, temperature, etc.
 - For example:
 - **Continuous (Uncountable) Infinite Example:**
 - **YouTube watch time in a Day $\rightarrow \Omega = \{x | x \in \mathbb{R}, 0 \leq x \leq 24\}$**

1.3.3 Elements of Probabilistic Space.

- A probability space is a mathematical triplet $(\Omega, \mathcal{F}, P)$ that provides the formal framework for probability theory.
- 2. **Event in a Sample Space (F) :**
 - An **event F** in a **sample space S(or Ω)** is defined as any **subset** of the **sample space**. i.e. **$F \subseteq S$**
 - Events represent outcomes or groups of outcomes that we are interested in observing.
 - **Events in Discrete Sample Spaces:**
 - **Simple Event:** An event that consists of **exactly one outcome** from the **sample space**.
 - **Example: Single coin flip $\rightarrow S = \{H, T\}$, $\leftarrow$ Outcomes**
 - **Event F_1 : Getting a Head $\rightarrow F_1 = \{H\}$**
 - **Events in Continuous Sample Spaces:**
 - **Example: You Tube watch time in a day: $S = \{x | x \in \mathbb{R}, 0 \leq x \leq 24\}$**
 - **Event $F_{\text{wasted day}}$: Watched You Tube for 5 or more hours. $\rightarrow F_{\text{wasted day}} = \{x | x \in \mathbb{R}, 5 \leq x \leq 24\}$**
 - This is a continuous event **over an interval**.
 - **Events as a Sets of Functions:**
 - Sometimes, especially in **infinite or functional sample spaces**, outcomes can be **functions**.
 - **Example: Toss a fair coin infinitely many times. $S = \{f: \mathbb{N} \rightarrow \{H, T\}\}$**
 - Each **function f** represents **one infinite sequence** of coin toss outcomes.
 - **Event F:** The first two tosses are tails: **$F = \{f \in S | f(1) = T \text{ and } f(2) = T\}$**
 - This is a functional event **defined** on the **behavior of a function over its domain**.

1.3.3.1 Events that are not Simple.

Complex Events

- A **complex event** refers to an event composed of
 - multiple simple events combined using logical operators such as
 - union ($\cup$), intersection ($\cap$), or complement (A').
 - It may involve **more than** one event but does not
 - specifically require their **simultaneous occurrence**.
- For example – Rolling a die:
 - Event A:** Rolling an even number:
 - $F_A := \{2, 4, 6\}$.
 - Event B:** Rolling a number greater than 4:
 - $F_B := \{5, 6\}$.
 - Complex Event:** Rolling an even number **or** a number greater than 4 :
 - $(F_A \cup F_B) = \{2, 4, 5, 6\}$

Joint Events

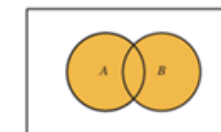
- A **joint event** refers to the **simultaneous occurrence** of **two or more events**.
- It is specifically concerned with their **intersection** ($A \cap B$) i.e., the **outcomes common to both events**.
- For example – Rolling a die:
 - Event A:** Rolling an even number:
 - $F_A := \{2, 4, 6\}$.
 - Event B:** Flipping heads on a single coin:
 - $F_B := \{H\}$.
 - Joint Event:** Rolling an even number and Flipping a head.
 - $(F_A \cap F_B) = \{(2, H), (4, H), (6, H)\}$.

In Figure →

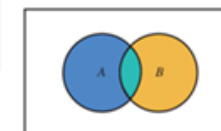
- not A $\{A^c\}$
- A or B $\{A \cup B\}$
- A and B $\{A \cap B\}$
- Disjoint.



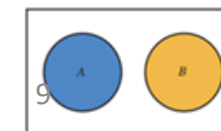
(a) Gold region = not A



(b) Gold region = A or B



(c) Green region = A and B



(d) Two disjoint events

1.3.4 Elements of Probabilistic Space.

- A **probability space** is a **mathematical triplet** $(\Omega, \mathcal{F}, \mathbb{P})$ that provides the formal framework for probability theory.

3. Probability Measure :

- A **probability measure** is a function that maps an events in an **event space** ($\mathcal{F}$) to a real number $[0,1]$ satisfying the **axioms of probability**:
 - $\mathbb{P}: \mathcal{F} \rightarrow [0, 1]$

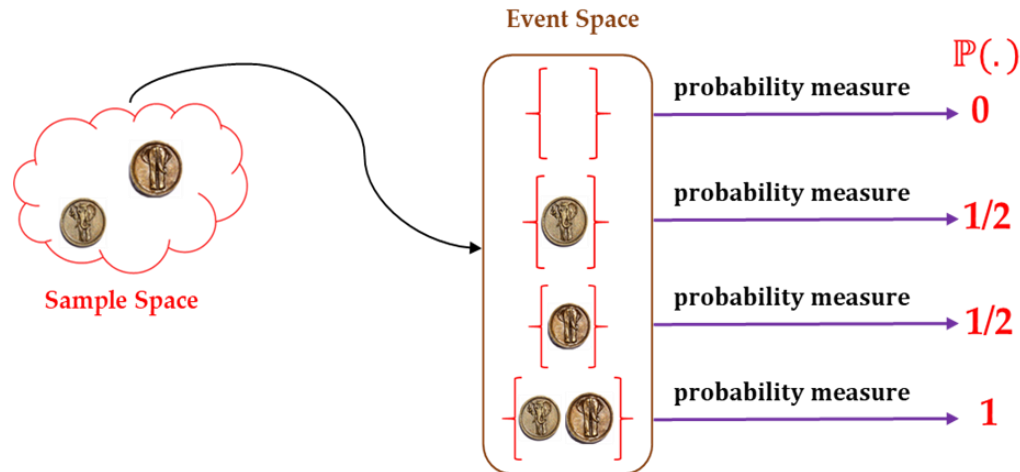


Fig: A Probabilistic Space.

←In a Figure:

Let $(\mathcal{S}, \mathcal{F}, \mathbb{P})$ be a probability space, where
 $\mathcal{S}$: Sample space (set of all possible outcomes).

$\mathcal{F}$: Event space.

$\mathbb{P}: \mathcal{F} \rightarrow [0, 1]$: A probability measure.

1.4 Kolmogorov's Three Axioms of Probability.

- Let S be the sample space, and let A be any event in S . A probability function P assigns a number to each event A such that:
 - Axiom 1: Non – Negativity**
 - For any event A , $P(A) \geq 0$
 - Interpretation: Probabilities are never negative. You cannot assign a negative likelihood to an event.
 - Axiom 2: Normalization (or Unit Measure)**
 - For the entire sample space S , $P(S) = 1$
 - Interpretation: Something from the sample space must happen – **the total certainty is 100%**.
 - Axiom 3: Additivity (for disjoint events)**
 - For two mutually exclusive (disjoint) events A and B , $P(A \cup B) = P(A) + P(B)$
 - Interpretation: If two events cannot happen at the same time (**disjoint or mutually exclusive events**), the probability that either occurs is the sum of their probabilities.
 - This extend to **any finite or countably infinite number of disjoint events** i.e.
 - For any sequence of disjoint or mutually exclusive events:
 - $A_1, A_2, \dots$ are $A_i \cap A_j = \emptyset$ for $i \neq j$; Then $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$.

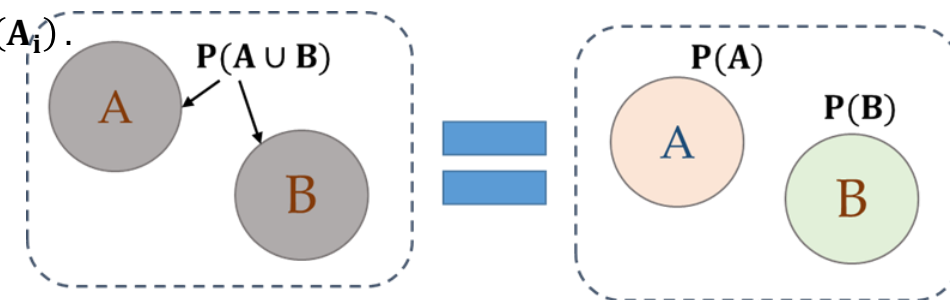


Fig: Understanding Additivity.

2. Understanding **Conditional Probability & Independence:** Quantifying **Dependent and Independent Events.** {Conditional Probability and Examination of Independence.}

2.1 Events that are Joint !!!

- **Joint Events and Joint Probability:**
 - Let **A and B** be two events defined on a **common sample space Ω** .
 - A **joint event** refers to the occurrence of both events **A and B**. The Probability of their simultaneous occurrence is denoted as:
 - **$P(A \cap B)$ { also written as $P(A, B)$ } = Probability both A and B Occur.**
 - and represents **the likelihood of both A and B occurring together.**
 - This is known as the **Joint Probability of A and B.**
 - This is useful, but **doesn't yet tell us how A and B are related.**
 - Example:
 - Event A: It rains today & Event B: The Traffic is heavy
 - Joint Event $A \cap B$: It rains and traffic is heavy.
 - We are now interested in not just single outcomes, but how two things occur together.
 - **Q: "When Events Occur Together: Are they Linked?"**
 - For example:
 - **If I told you it's raining, how likely is it that traffic is heavy?**
 - **Do you think rain and traffic always happen together? Could one cause the other?**

2.1.1 Understanding Dependence and Independence.

- Q: “When Events Occur Together: Are they Linked?”
 - Do you think rain and traffic always happen together? Could one cause the other?
 - While Computing joint probabilities is useful, it is often more important to understand the relationship between events:
 - Are they related in any meaningful way?
 - Does the occurrence of one event influence the likelihood of the other?
 - This leads to two important types of relationships between events:
- **Dependent Events:**
 - Events A and B are said to be dependent if the occurrence of one affects the probability of the other.
 - That is: $P(A|B) \neq P(A)$.
- **Independent Events:**
 - Events A and B are independent if knowledge of one event provides no information about the other.
 - Mathematically they satisfy: $P(A \cap B) = P(A) \cdot P(B)$ or equivalently $P(A|B) = P(A)$

2.1.2 Exploring Dependent Events.

- Simple (or marginal) probability is **inadequate** when the occurrence of one event **influences** the likelihood of another — a situation known as **event dependence**.
 - In such cases, treating events as if they were independent can lead to **incorrect inferences**.
 - We need a **framework** that **accounts** for **how new information alters our belief** about another event.
- This is where **Conditional Probability** comes in.
- **Example: Weather Forecasting**
 - Q: *“What is the probability that it will rain, given that the sky is cloudy?”*
 - We can not simply compute $P(\text{Rain})$ in isolation, because:
 - The information that “**it is cloudy**” provides **additional knowledge**.
 - This change our belief about likelihood of rain.
 - Thus, we must compute the conditional probability:
 - **$P(\text{Rain}|\text{Cloudy})$**
 - This **accounts** for the **dependence** between **cloudiness and rain**, and **updates our estimate** accordingly.
- **What is Conditional Probability?**

2.2 Conditional Probability: Updating Beliefs with Evidence.

- **Theory of conditional probability** provides the way to measure the probability when two events are dependent with each other, i.e. one event can only occur if another event has already happened.
- For any event A and B, if event A is dependent on B (or also called conditioned on B) and we know $P(B) > 0$ Then **Conditional Probability of A given B** is :
 - $P(A|B) = \frac{P(A \cap B)}{P(B)}$ is called conditional probability of A conditioned on B.
- **Relative Frequency Interpretation:**
 - Suppose we repeat the experiment many times.
 - Out of all trials, the fraction in which B occurs is approximately $P(B)$.
 - Out of those where B occurs, the fraction in which both A and B occur is approximately $P(A \cap B)$.
 - So , the fraction of times that A occurs within the occurrence of B is:
 - $P(A|B) = \frac{\text{\#count both A and B occur}}{\text{\#count B occur}} == \frac{P(A \cap B)}{P(B)}$, if $P(B) > 0$

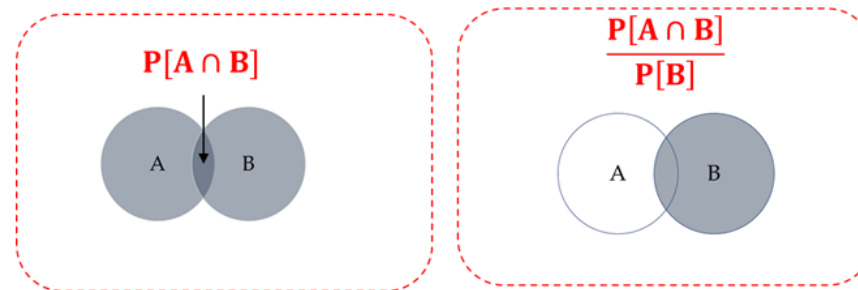
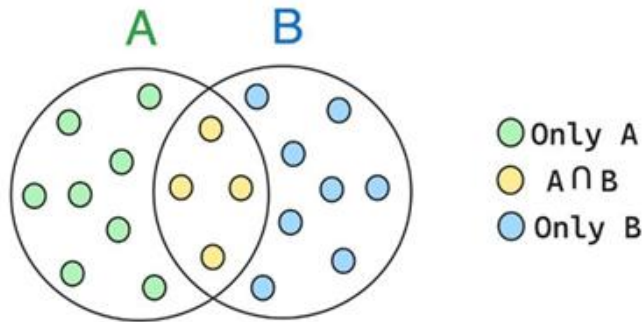


Fig: Illustration of Conditional Probability and comparison to Joint Probability.

2.2.1 Conditional vs. Joint Probability

- Conditional probability refines **our estimate of an event's likelihood based on known evidence**
 - while joint probability **quantifies the simultaneous occurrence of two events in the full context.**

Joint vs Conditional Probability!



Joint probability

Case1: A & B are independent

$$P(A \cap B) = P(A) \times P(B)$$

Case2: A & B are not independent

$$P(A \cap B) = P(A) \times P(B|A)$$

Probability of two events happening simultaneously

Conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability that A occurs given that B has already occurred

Image by: akshay_pachar

Key Distinction: Conditional vs. Joint Probability.

Concept	Formula	Focus
Conditional Probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$	- Focuses on the likelihood of A, given that B has occurred. Conditional Probability isolates the impact of B on A, focusing only on cases where B occurs.
Joint Probability	$P(A \cap B)$	- Measures the Probability that both A and B Occur together in the full sample space. Joint Probability measures the overlap of A and B in the entire sample space.

2.2.2 Axioms of Conditional Probability.

- **Conditional probability defines a new probability measure** over the sample space,
 - where **probabilities are scaled relative to the occurrence of B** and are bounded by following **axioms**:

Axiom	Mathematical Expression	Interpretations
Non - Negativity	$P(A B) \geq 0$	The probability of any event A, given B, is always non negative.
Normalizations	$P(\Omega B) = 1$	Given B has occurred, some outcome in the sample space must occur.
Additivity	$P(A_1 \cup A_2 \cup \dots B) = \sum P(A_i B)$	If events $A_1, A_2, \dots$ are disjoint , the conditional probability of their union equals the sum of their individual conditional probabilities.
Complement	$P(A^c B) = 1 - P(A B)$	Complement of conditional Probability.

- These axioms ensure that **conditional probability behaves like a proper probability measure**, but within the **restricted sample space defined by event B**.

2.3 When Events Don't Influence Each Other: Independence

- We have introduced the conditional probability for event A conditioned on event B $\{P(A|B)\}$ to capture the partial information that event B provides about event A.
 - An interesting and important special case arises when the occurrence of B provides **no information** and does not alter the probability that A will occur, i.e. when two events are independent:
 - Two events are said to be **independent** if knowing the outcome of one event does not change your belief about whether or not the other event will occur. Mathematically:
 - $P(A \cap B) = P(A) \times P(B)$ {aka multiplicative rule of probability}
- Example Scenario – Coin Tosses:
 - Suppose you toss a coin twice:
 - $A = \{\text{Heads on the first toss}\}$,
 - $B = \{\text{Heads on the second toss}\}$.
 - What is the probability of getting A and B:
 - Since the outcome of the first toss does not affect the second:
 - $P(A \cap B) = P(A) \times P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

2.4 Chain Rule of Probability.

- The chain rule of probability express the probability of the intersection of multiple events in terms of conditional probabilities. For n events $A_1, A_2, \dots, A_n$ the chain rule is:
 - $P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2) \dots P(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1})$
- This rule allows us to compute the joint probability of multiple events step by step, starting from the probability of the first event and progressively conditioning on previous events.
- **Independent Events in the Chain Rule:**
 - If the n events $A_1, A_2, \dots, A_n$ are independent then, chain rule simplifies to
 - $P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2) \dots P(A_n) = \prod_{i=1}^n P(A_i)$

3. Understanding Bayes Theorem.

{Deriving Probabilistic Reasoning from Conditional Probability.}

3.1 Challenge with Conditional Probability.

- **The core Problem:**
 - We often want to know how likely something is, given some observed evidence.
 - **Example 1: Medical Diagnosis**
 - Let:
 - **$D \rightarrow$ Patient has Disease.**
 - **$T \rightarrow$ Test comes back positive.**
 - Doctors want to know:
 - What is the chance the patient is sick if the test is positive?
 - **$P(\text{Disease}|\text{Positive Test}) = \frac{P(\text{Disease} \cap \text{Positive Test})}{P(\text{Positive Test})}$;**
 - Looks Simple, so what is the problem?
- **But in most real – world situations:**
 - We do not directly observe the probability of the cause.
 - We do observe the effect (evidence) and know how likely that effect is under different causes.

3.1.1 Challenge with Conditional Probability.

- Why We *Don't* Know $P(\text{Disease} \cap \text{Positive Test})$ Directly?
 - Even though $P(\text{Disease} \cap \text{Positive Test})$ (i.e. the joint probability of having the disease and testing positive) is conceptually simple,
 - here is why it is typically not directly available.
 - We test people, But Don't know the True Condition:
 - In real populations:
 - You can observe who tested positive from Test Result.
 - But you often do not know who truly has the disease unless you use a perfect Test,
 - i.e. Test with 100% accuracy.
 - These Test may not exist or may be very expensive/invasive not feasible.
 - That means, we can count positives, but not confidently count true positives $\rightarrow$ so we can not directly observe the joint events.
- Then what do we do in Practice?

3.1.2 Challenge with Conditional Probability.

- But in Practice we do have:
 - **P(Positive Test|Disease) → likelihood:**
 - Observed from controlled clinical Studies (e.g. Clinical Trial)
 - Example: Out of 8,000 people who had the disease, 7,999 tested positive.
 - This gives us the sensitivity of the test.
 - **P(Disease) → Prior:**
 - This is the prevalence of the Disease in real population and represents how common the disease is in the general population
 - which can be observed through large scale epidemiological studies.
 - **P(Positive Test) → Evidence:**
 - This can be computed using Law Total probability.

$$P(\text{Positive Test}) = P(\text{Positive Test}|\text{Disease}) \cdot P(\text{Disease}) + P(\text{Positive Test}|\text{No Disease}) \cdot P(\text{No Disease})$$

- **P(Disease ∩ Positive Test)** could not be measured directly but now could be reconstructed using chain rule of Probability i.e.

$$P(\text{Disease} \cap \text{Positive Test}) = P(\text{Positive Test}|\text{Disease}) \cdot P(\text{Disease})$$

- Based on all above observation, Bayes Rule {or, Theorem} flips the conditional and lets you compute:

$$P(\text{Disease}|\text{Positive}) = \frac{P(\text{Disease} \cap \text{Positive})}{P(\text{Test Positive})}$$

$$P(\text{Disease}|\text{Positive}) = \frac{P(\text{Positive Test}|\text{Disease}) \cdot P(\text{Disease})}{P(\text{Test Positive})}$$

3.2 Bayes Theorem: Formal Definition.

- Bayes' rule states:

$$\text{posterior} \propto \text{likelihood} \times \text{prior} \Rightarrow \text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}$$

- Bayes' Rule allows us to *update our beliefs* about *the probability* of *an event* by incorporating *new information* or *evidence*, making it a powerful tool for *adaptive decision-making*.
 - This "*updating*" process is fundamental in probabilistic reasoning and plays a key role in fields like medicine, machine learning, and finance.
- Let's break down *how this updating process works with Bayes' Rule* based on following scenario:
- Cautions!!!:
 - **Bayes Rule or Theorem is not a new formula;**
 - it is a **re – expression of conditional probability** that:
 - Uses known quantities,
 - Allows us to reason from effect back to cause,
 - Enables belief updating in face of new evidence.

3.3 Understanding Terminologies in Bayes Rule.

- **Example – 2 – Real world – Fraud Detection:**

- We want to know: What is the probability a transaction is fraudulent, given that triggered an alert?

- **Posterior $\Rightarrow P(\text{Fraud}|\text{Alert})$**

- We know we can not observe, **$P(\text{Fraud} \cap \text{Alert})$** as we can not directly count

- **how many transactions are both fraudulent and triggered an alert.**

- But we can observe:

- **Likelihood $P(\text{Alert}|\text{Fraud})$** – the probability that a fraudulent transaction triggers an alert,

- **Prior $P(\text{Fraud})$** – the base rate or prevalence of fraud in population.

- **Evidence $P(\text{Alert})$** – the overall probability that any transaction triggers an alert.

- **$\{\text{posterior}\} P(\text{Fraud}|\text{Alert}) = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}} = \frac{P(\text{Alert}|\text{Fraud}) \times P(\text{Fraud})}{P(\text{Alert})} = ?$**

- **Question :**

- How can **each of the three components** above (likelihood, prior, and evidence) be **estimated or obtained in practice** in a **real – world fraud detection system**?

What we Can Observe and Estimate:

Component	Meaning	How it's Observed	Suppose
 Likelihood: $P(\text{Alert} \text{Fraud})$	The chance that a fraudulent transaction triggers an alert.	Estimated using labeled data from past confirmed fraud cases (e.g., from investigations or audits)	$P(\text{Alert} \text{Fraud}) = 0.98$ {High detection rate}
 Prior $P(\text{Fraud})$	The overall rate of fraudulent transactions	Estimated from historical transaction data (e.g. 0.5% of all transaction flagged in past were confirmed fraud)	$P(\text{Fraud}) = 0.005$ {0.5% of all transactions are fraudulent}
 Evidence $P(\text{Alert})$	The overall probability that any transaction triggers an alert	Computed from operational logs of the fraud detection system (e.g. 2% of transactions trigger alerts)	$P(\text{Alert}) = 0.02$ {2% of all transactions trigger alerts}

3.4 Putting It All Together

$$P(\text{Fraud}|\text{Alert}) = \frac{P(\text{Alert}|\text{Fraud}) \times P(\text{Fraud})}{P(\text{Alert})} = \frac{0.98 \times 0.005}{0.02} = \frac{0.0049}{0.02} = 0.245 = 24.5\%$$

Interpretations:

- So, **only 24.5%** of transactions that trigger an alert are **actually fraudulent** — despite a high detection rate.

Key Insight:

- Even with a **strong alert system**, if **fraud is rare**, **most alerts** may still be **false positives**.
- **Bayes Rule** helps us **quantify this uncertainty** and make **informed decisions** (e.g. **whether to investigate**).

3.5 Summarizing Bayes Rule!!!

- The Common Problem we face:
 - We often want to answer questions like:
 - *“Given this evidence, what is the probability that a particular hypothesis is true?”* This is: $P(H|E) = ?$
 - We do not have direct access to joint probability of hypothesis and evidence i.e. $P(H \cap E)$.
- What we usually know or we can measure or estimate are likelihood, the prior, and the evidence,
 - and apply a bayes rule such that:

Likelihood:

- How probable the evidence is, assuming the hypothesis is true.
- e.g., If a transaction is actually fraud, how likely is it to trigger an alert?

Prior:

- How probable the hypothesis is before seeing any evidence.
- How common is fraud in general?

$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$

Posterior:

- How probable the hypothesis is, assuming the evidence is observed.
- e.g., If a transaction triggered an alert, how likely is it to be fraud?

Evidence:

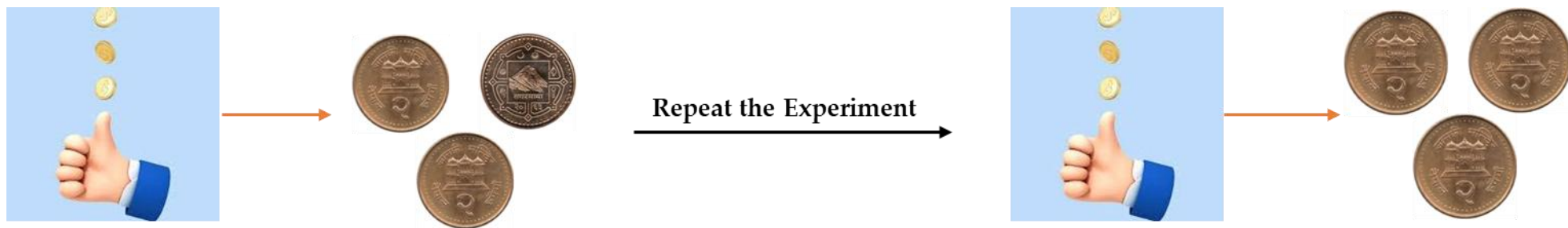
- How probable the evidence is, considering all possible causes.
- What is the overall chance that any transaction (Fraudulent or Not) triggers an alert?
- A Total Probability.

4. Transition.

{From Traditional Probabilistic Thinking to ML Oriented Thinking.}

4.1 Traditional Probabilistic Thinking.

- **Probability of Events — Yes, or No?**
 - We've been answering questions that deal with **binary events** — things that either happen or don't. For e.g.,:
 - “What is the probability that a user clicks the email?”
 - “What is the probability of getting 2 heads in 3 tosses?”
 - “What is the probability that a transaction is fraudulent given it triggered an alert?”
 - These questions help us understand **likelihood**, **update beliefs**, and **make decisions under uncertainty**.
 - But they all focus on *whether* something happens — **not** *how much* or *how often*.
 - **Now We Want to Go Further: Quantity Matters**



4.2 ML Oriented Example.

- **For the task of House Price Prediction:**
 - Traditional Probability could only answer the Question:
 - *“What’s the probability that house price is above \$300K?”*
 - This treats the outcome as a yes/no event, like in coin tosses or dice rolls.
- **What actually we are interested on (ML oriented thinking):**
 - *“Given a house with features x_1, x_2, x_3 , what price will it likely sell for?”*
 - This is not the binary event,
 - **we care about a numerical output, and that output varies even for similar features.**
 - So, we ask:
 - What value will the prices produce?
 - What is the distribution of possible values?
 - What is the average price, variance etc.
- “So far, we’ve studied the *probability of events*. But when we care about *values*, not events, we need a new tool:
 - **the random variable.”**

4.3 So, What are Random Variables?

- In simple terms, a **random variable** is a **numerical value** $\mathbb{R}$ assigned to each outcome of a random or chance experiment.
- **Formal Definition:**
 - A **random variable** is a **function** that **maps** outcomes of a **sample space** to **real numbers**:
 - $\mathbf{X: \Omega \rightarrow \mathbb{R}}$
 - That means for each **outcome** $\omega \in \Omega$, the **random variable** assigns a **single numeric** value:
 - $\mathbf{X(\omega) \in \mathbb{R}}$.
 - **Cautions:**
 - This **function must be measurable**,
 - meaning the **set of values**, it takes must align with **the structure of probability**
 - so, we can compute things like $\mathbf{P(X \leq x)}$.

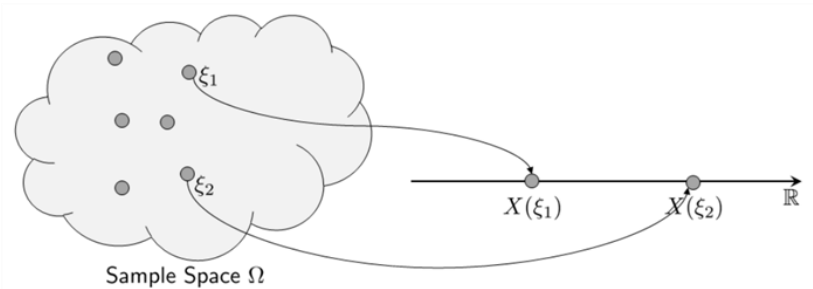


Fig: A definition of Random Variable.

To Summarize:

Concept	Description
Sample Space Ω	All possible outcomes of a random process.
Random variable X	A function assigning numbers to each outcome.
Output $X(\omega)$	A real number representing the outcome.

4.4 Types of Random Variables.

Discrete Random Variable

- A **discrete random variable** is one that can take on a **countable number of distinct values**.
 - Often represents **counts** or **categories**.
 - Values can be listed individually (finite or countably infinite).
- **Examples:**
 - **Number of emails clicked in a campaign**
 - **Number of customers arriving per hour**
 - **Outcome of a dice roll $X \in \{1, 2, 3, 4, 5, 6\}$**
- **Formal:**
 - **$X: \Omega \rightarrow \{x_1, x_2, x_3, \dots\}$, where each $x_i \in \mathbb{R}$**

Continuous Random Variable

- A **continuous random variable** can take on any **real value** within **an interval** (or union of intervals).
 - Often represents measurements like time, distance, weight, or revenue
 - Values are uncountably infinite
- **Examples:**
 - **Time until next customer arrives**
 - **Revenue generated by a customer**
 - **Temperature at noon**
- **Formal:**
 - **$X: \Omega \rightarrow \mathbb{R}$, and $P(X = x) = 0$ for any **single x** .**
 - Instead, we compute probabilities over intervals,
 - like **$P(a \leq X \leq b)$** .

4.5 Probability Distribution Function.

- A **Probability Distribution Function** describes how probability is distributed over the possible values of a random variable whether **discrete or continuous**.
- In General Terms:
 - A **Probability Distribution Function** tells us:
 - What values a random variable can take.
 - And how likely each value (or range of values) is to occur
- It can be represented in Different Forms:
 - **Table:** A list of values and their probabilities.
 - **Graph:** Bar chart for discrete, smooth curve for continuous
 - **Formula/Equation:** Mathematical expressions be like $P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$

4.5.1 PDF: Discrete vs. Continuous.

For Discrete Random Variables

- We use the **Probability Mass Function (PMF)**
- It gives the probability of each **countable outcome**:
 - $P(X = x_i)$
 - $\sum_i P(X = x_i) = 1$

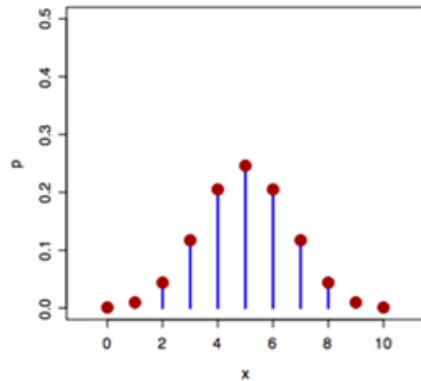


Fig: Discrete Probability Distribution

For Continuous Random Variables

- We use the **Probability Density Function (PDF)**.
- Since continuous values are uncountable,
 - the probability of any exact value is
 - $P(X = x) = ?$
- Instead, we calculate the probability over an interval:
 - $P(a \leq X \leq b) = \int_a^b f(x) dx$

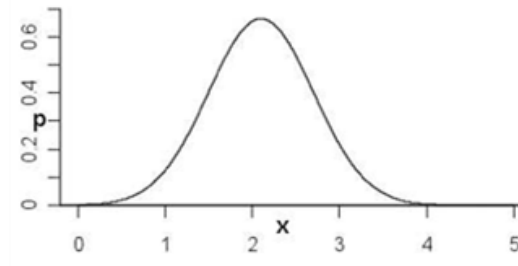


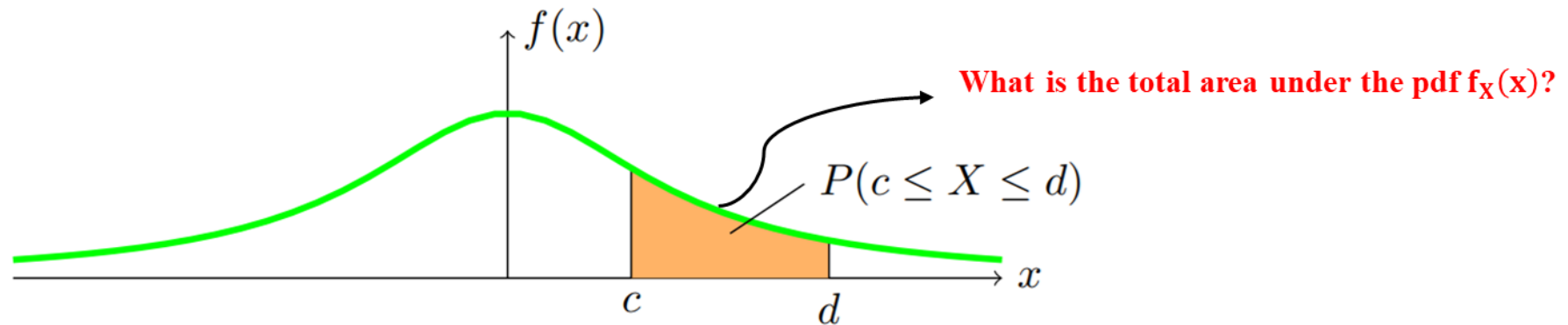
Fig: Continuous Probability Distribution
image from internet : may subjected to copyright.

4.6 PMF: A General Definition.

- A **Probability Mass Function (PMF)** is a function that describes the **probability of each possible value** that a **discrete random variable** can take.
- In simple terms:
 - A **PMF** tells us:
 - “What values the variable can take: e.g. $\{0, 1, 2, \dots\}$ ”
 - “And how likely each of those values is to occur”
- Formally:
 - For a **discrete random variable X** , the **PMF** is a **function**:
 - **$P(X = x) = p(x)$** such that:
 - **$p(x) \geq 0$ for all $x \in \text{Range}(X)$**
 - **$\sum_x p(x) = 1$**
- Representations:
 - A **PMF** can be expressed as:
 - A **table** listing values and their probabilities.
 - A **graph (bar chart)** showing probabilities visually.
 - A **formula or mathematical equations**.

4.7 Continuous Random Variable

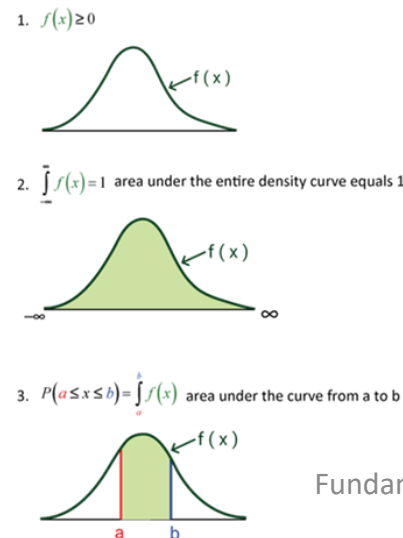
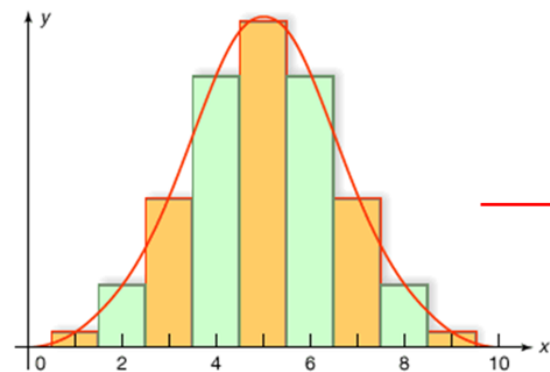
- A **continuous random variable** takes a range of values, which may be finite or infinite in extent.
 - Here are a few examples of ranges: $[0, 1]$, $[0, \infty)$, $(-\infty, \infty)$, $[a, b]$.
- Formal Definition:
 - A **random variable** X is **continuous** if there is a function $f_X(x): \mathbb{R} \rightarrow [0, \infty)$ such that for **any** $a \leq b$ we have:
 - $P(a \leq X \leq b) = \int_a^b f_X(x) dx$
 - The **function** f_X is called the **probability density function (pdf)** of X .
 - If you graph the **probability density function** of a **continuous random variable** X then:



$P(c \leq X \leq d) = \text{area under the graph between } c \text{ and } d.$

4.7.1 What is Probability Density Function (PDF)?

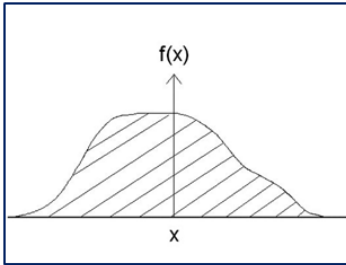
- A **PDF** is a **function** $f_X(x)$ that satisfies all the axioms of probability written as:
 - For an **interval** $[a, b]$, the probability is: $P(a \leq X \leq b) = \int_a^b f_X(x) dx$
 - Axioms of Probability:**
 - Non-negativity:** integration over any region of the sample space must never produce a negative value: $f_X(x) \geq 0 \forall x \in \mathbb{R}$
 - Exhaustive:** Over the whole sample space, the probability function must integrate to one i.e. the total area under the curve is 1: $\int_{-\infty}^{\infty} f_X(x) dx = 1$
 - Additive:** The probability of the union of any non-overlapping regions is the sum of the individual regions.



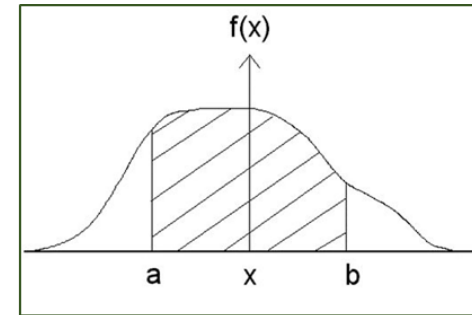
Feature	Discrete PMF	Continuous PDF
Values	Countable	Uncountably infinite
Probability at point	$P(X = x) > 0$	$P(X = x) = 0$
Total probability	$\sum P(X = x) = 1$	$\int f(x) dx = 1$
Plot type	Bar Chart	Smooth Curve
Computation	Exact values	Area under curve over interval.

Table: PMF vs. PDF

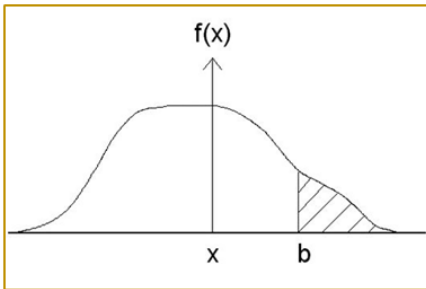
4.7.2 Graphical view of PDF.



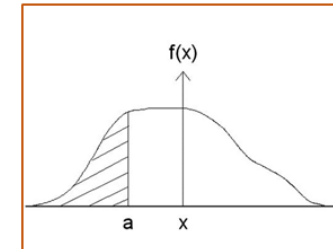
$$\text{Total area} = \int_{-\infty}^{\infty} f_X(x) dx = 1$$



$$\text{Area } P[X \in [a, b]] = \int_a^b f_X(x) dx$$



$$\text{Area } P_X(x \geq b) = \int_b^{\infty} f_X(x) dx$$



$$\text{Area } P_X(x \leq a) = \int_{-\infty}^a f_X(x) dx$$

4.8 Expectation of Discrete Random Variable

- A random variable is fully represented by its probability mass function (PMF), which represents each of the values the random variable can take on, and the corresponding probabilities.
- Expectation can be thought as a {**Summary statistics**} - - **Weighted Average** of a Random Variable!

Definition: Expectation

The expectation of a {discrete} random variable Y , written $E[Y]$ is the **average** of all the values the random variable can take on, each **weighted** by the **probability** that the random variable will take on that value:

$$E[Y] = \sum_y y \cdot p(Y = y)$$

Expectation goes by many other names such as:

Mean, weighted Average, Center of Mass, 1st Moment.

If $p(y)$ is an **accurate characterization** of the **population frequency distribution**, then $E[Y] = \mu$ is the **population mean**.

4.8.1 Properties of Expectations.

- **Property: Linearity of Expectation**

- $E[aY + b] = aE[Y] + b$
 - Where a and b are constants and not random variables.

- **Property: Expectation of Constant**

- $E[a] = a$
- Sometimes in proofs, you will end up with the expectation of a constant (rather than a random variable). For example, what does the $E[5]$ mean?
 - Since 5 is not a random variable, it does not change, and will always be 5, $E[5]=5$

- **Property: Law of Unconscious Statistician**

- $E[g(X)] = \sum_x g(x)P(X = x)$
- One can also calculate the expected value of a function $g(X)$ of a random variable X when one knows the probability distribution of X but one does not explicitly know the distribution of $g(X)$.
- This theorem has the humorous name of "the Law of the Unconscious Statistician" (LOTUS), because it is so useful that you should be able to employ it unconsciously.

4.8.2 Properties of Expectations.

- **Property: Expectation of the Sum of Random Variables**

- $E[X + Y] = E[X] + E[Y]$
- This is true regardless of the relationship between X and Y . They can be dependent, and they can have different distributions. This also applies with more than two random variables.
- $E[\sum_{i=1}^n X_i] = \sum_{i=1}^n E[X_i]$
- **Sample – Proof:**

$$\begin{aligned} E[X + Y] &= \sum_{x,y} (x + y)P(X = x, Y = y) \quad \{\text{Expected value of a sum.}\} \\ &= \sum_{x,y} [xP(X = x, Y = y) + yP(X = x, Y = y)] \quad \{\text{Distributive Property of sums.}\} \\ &= \sum_{x,y} xP(X = x, Y = y) + \sum_{x,y} yP(X = x, Y = y) \quad \{\text{Commutative property of sums.}\} \\ &= \sum_x \sum_y xP(X = x, Y = y) + \sum_x \sum_y yP(X = x, Y = y) \quad \{\text{Expanding Sums.}\} \\ &= \sum_x x \sum_y P(X = x, Y = y) + \sum_y y \sum_x P(X = x, Y = y) \quad \{\text{Distributive property of sums.}\} \\ &= \sum_x xP(X = x) + \sum_y yP(Y = y) \quad \{\text{Marginalization.}\} \\ &= E[X] + E[Y] \quad \{\text{Definition of Expectation.}\} \end{aligned}$$

4.8.3 Variance of Discrete Random Variable.

- In the last slide we showed that Expectation was a useful summary of a random variable
 - it calculates the "**weighted average**" of the **random variable**.
- One of the next most important properties of random variables to understand is variance:
 - **the measure of spread**

Definition Variance:

The **variance** is a measure of the "**spread**" of a **random variable** around **the mean**.

Variance for a random variable, **Y**, with expected value $E[Y] = \mu$ is:

$$\text{Var}(Y) = E[(Y - \mu)^2]$$

Semantically, this is the **average distance of a sample from the distribution to the mean**.

When computing the variance often we use a different (equivalent) form of the variance equation:

$$\text{Var}(Y) = E[Y^2] - E[Y]^2$$

4.9 Expectation and Variance of CRV.

- **Expectation of a Continuous Random Variable:**

- For a *continuous random variable* X with **pdf** $f_X(x)$ we define the **expectation** $\mathbb{E}[X]$ as:

- $\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$

- **Condition for Existence:**

- The expectation exists (i.e. finite) if and only if:

- $\int_{-\infty}^{\infty} |x| \cdot f_X(x) dx < \infty$

- If the above **integral diverges** (i.e. equals $+\infty$), then the **expectation is not defined**.

- **Interpretation:**

- The expectation is a **weighted average of all possible values of X** , where the weights are given by **the density $f_X(X)$** .
- It gives us the **center of mass** of the distribution curve.

4.9 Expectation and Variance of CRV.

- **Variance of a Continuous Random Variable:**

- For a *continuous random variable* X with **pdf** $f_X(x)$ and expectation $\mathbb{E}[X]$, the **variance** is defined as:

- $\text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f_X(x) dx$

- where $\mu = \mathbb{E}[X]$ is the **expected value** mean of X .

- If the **expectation** $\mathbb{E}[X^2]$ **exists**, variance can also be computed as:

- $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

- **Interpretation:**

- Variance **measures** the **spread or dispersion** of a **distribution** around its mean.
 - A **larger variance** implies the **values of X** are **more spread out** from the mean.

4.10 But What if we Ask ...

- “What’s **the probability** that a **batch** gets **at most 1 click** ?”
 - $P(X \leq 1) = ?$
 - PMF gives values only at individual points.
 - To answer questions involving ranges or cumulative behavior,
 - PMF becomes tedious and error prone, **especially for large outcome spaces**.
- **CDF to the Rescue:**
 - The **cumulative Distribution Function (CDF)** gives us:
 - $F_X(x) = P(X \leq x) = P(X = 0) + P(X = 1) = 0.54$.

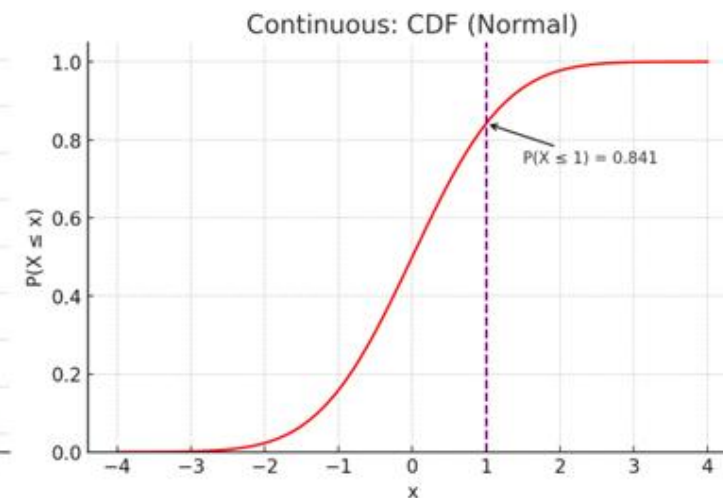
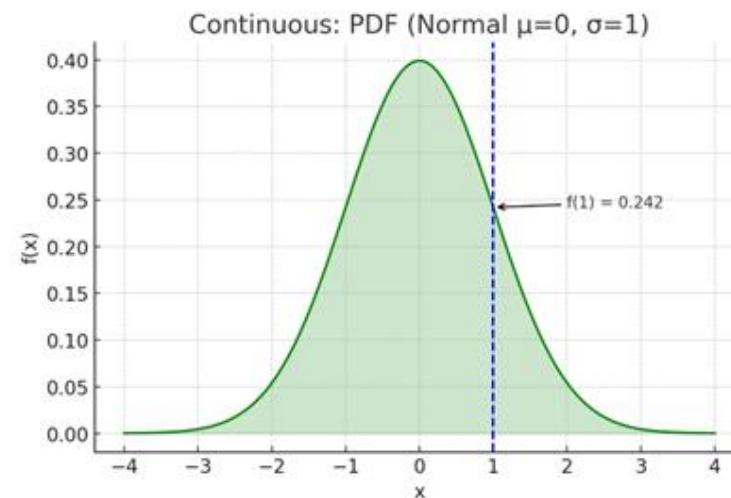
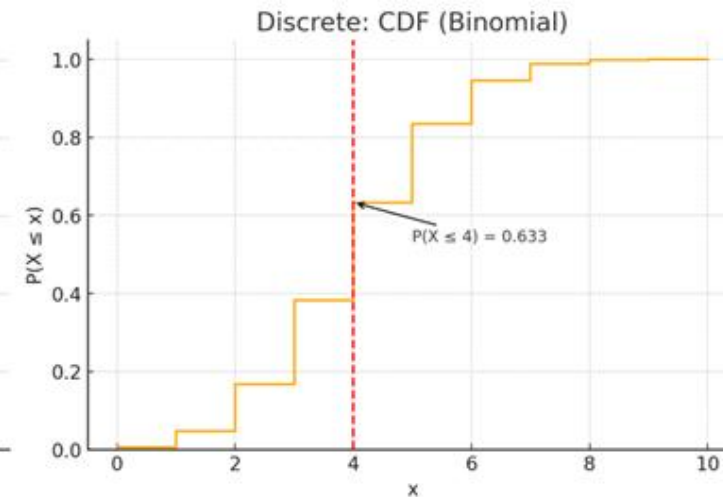
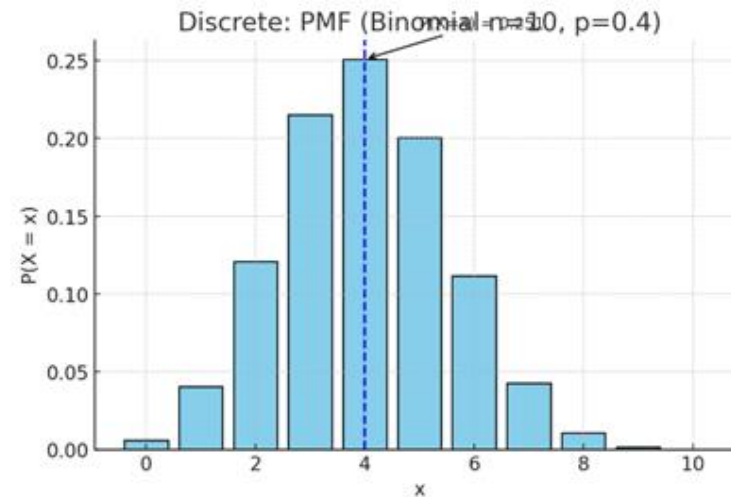
Table: Possible CDF for our Scenario.

Clicks in Batch	Frequency	$\hat{P}(X = x)$	$F_X(x) = P(X \leq x)$
0	90	$\frac{90}{500} = 0.18$	0.18
1	180	$\frac{180}{500} = 0.36$	$0.18 + 0.36 = 0.54$
2	120	$\frac{120}{500} = 0.24$	$0.54 + 0.24 = 0.78$
3	110	$\frac{110}{500} = 0.22$	$0.78 + 0.22 = 1.00$

4.11 Definition: Cumulative Distribution Function.

- For a *random variable* X , the cumulative distribution function (cdf), **denoted by $F_X(x)$** , is defined as:
 - **$F_X(x) = P(X \leq x)$**
 - which means the probability that the **random variable X** takes on a value less than **or equal to x** .
 - For **discrete random variables**, the CDF is the **sum of probabilities** of all outcomes less than or equal to x :
 - **$F_X(x) = P(X \leq x) = \sum_{t \leq x} P(X = t)$**
 - For **continuous random variables**, the CDF is the **integral of the probability density function (PDF)** up to x :
 - **$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt$**
 - the area under the CDF between **$-\infty$ and x** .

4.12 Distribution and Cumulative Function.



Thank You.